Basic Concepts and Linear Algebra

Optimization Problem Standard Form

$$\begin{aligned}
\min/\max \quad & f(x) && (1)\\
\text{s.t.} \quad & g_i(x) \le b_i \quad \text{(inequality constraints)} && (2)\\
& h_j(x) = c_j \quad \text{(equality constraints)} && (3)\\
& x \in X \quad \text{(variable bounds)} && (4)
\end{aligned}$$

Linear Algebra Fundamentals

- **Rank**: Maximum number of linearly independent rows/columns
- **Matrix multiplication**: $C = AB$, $c_{ij} = \sum_k a_{ik} b_{kj}$
- **Matrix inverse**: $AA^{-1} = I$
- **System of equations** $Ax = b$:
  - $m = n$, $\text{rank}(A) = n \to$ unique solution
  - $m < n \to$ infinite solutions
  - $\text{rank}(A) < \text{rank}([A|b]) \to$ no solution

Gradient and Hessian

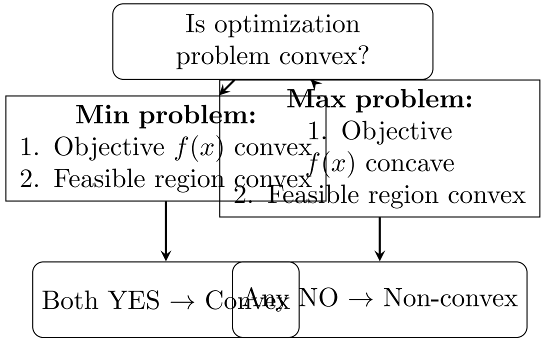
- **Gradient**: $\nabla f(x) = \begin{bmatrix} \frac{\partial f}{\partial x_1} \\ \vdots \\ \frac{\partial f}{\partial x_n} \end{bmatrix}$
- **Hessian**: $\nabla^2 f(x) = \left[\frac{\partial^2 f}{\partial x_i \partial x_j} \right]$
- **1st order Taylor**: $f(x) \approx f(x_0) + \nabla f(x_0)^T (x - x_0)$
- **2nd order Taylor**: $f(x) \approx f(x_0) + \nabla f(x_0)^T (x - x_0) + \frac{1}{2}(x - x_0)^T \nabla^2 f(x_0)(x - x_0)$

Optimality Conditions

- **First-order necessary**: $\nabla f(x^*) = 0$
- **Second-order sufficient**:
  - min problem: $\nabla^2 f(x^*)$ positive definite (PD)
  - max problem: $\nabla^2 f(x^*)$ negative definite (ND)

Convex Programming and Linear Programming

Convex Programming Decision



Linear Programming

Standard Form:

$$\begin{aligned}
\min \quad & c^T x && (5)\\
\text{s.t.} \quad & Ax = b && (6)\\
& x \ge 0 && (7)
\end{aligned}$$

Key Terms:

- **Basic solution**: Solution with n basic variables
- **Basic feasible solution (BFS)**: Basic + feasible
- **Degeneracy**: Basic variable = 0
- **Extreme point**: Vertex of polyhedron
- **Extreme ray**: Unbounded direction

Simplex Method Steps

1. Find **initial BFS**
2. **Optimality test**: Reduced costs ≤ 0
3. **Direction**: Choose entering variable
4. **Step size**: Minimum ratio test
5. **Update basis**: Choose leaving variable

Special Problems and Practical Formulas

Transportation Problem

$$\min \quad \sum_i \sum_j c_{ij} x_{ij} \tag{8}$$
$$\text{s.t.} \quad \sum_j x_{ij} \le s_i \quad \text{(supply)} \tag{9}$$
$$\sum_i x_{ij} \ge d_j \quad \text{(demand)} \tag{10}$$
$$x_{ij} \ge 0 \tag{11}$$

Infeasibility: $\sum s_i < \max\{d_j\}$

Absolute Value Linearization

$$\min |x-a| \quad \to \quad \min z \tag{12}$$
$$\text{s.t.} \quad z \ge x - a \tag{13}$$
$$z \ge -(x-a) \tag{14}$$

Set Properties

Set $X = \{x \in [a,b]^n : \sum x_i \ge c\}$

- **Bounded**: $[a,b]$ finite intervals
- **Closed**: Defined by equalities/inequalities
- **Convex**: Intersection of box + linear constraints

Common Convex Functions

- x^k ($k \ge 1$, $x \ge 0$): convex
- $\ln(x)$ ($x > 0$): concave
- e^x: convex
- $x^2 + y^2$: convex
- $\max\{f_1(x), f_2(x)\}$: convex if f_1, f_2 convex
- $|x|$: convex

Decision Timing

- **Here-and-now**: Decide before uncertainty
- **Wait-and-see**: Decide after uncertainty

Key Calculations from Past Exams

- **Gradient**: $f(x_1, x_2) = \sqrt{x_1 - 1} + (x_2 - 1)^{1/4}$ at $(3, 2)$

$$\nabla f = \begin{bmatrix} \frac{1}{2\sqrt{2}} \\ \frac{1}{4} \end{bmatrix}$$

- **Hessian**: $f(x_1, x_2) = 2(x_1 - 3)^2 + 6x_1x_2 + 2(x_2 + 3)^2 + 6x_1 + 2x_2$ at $(2, 3)$

$$H = \begin{bmatrix} 4 & 6 \\ 6 & 4 \end{bmatrix}$$

Important Abbreviations

- **LP**: Linear Programming
- **NLP**: Nonlinear Programming
- **BFS**: Basic Feasible Solution
- **PSD**: Positive Semidefinite
- **PD**: Positive Definite
- **s.t.**: subject to